IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MICROELECTROMECHANICAL SYSTEMS (MEMS)

Thursday, 7 May 2026, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer Question ONE, and any TWO of the remaining questions (Questions 2, 3, and 4).

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : Z. Durrani

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Information for Candidates

The following physical constants may be used:

Permittivity of free space: $\varepsilon_0 = 8.854 \times 10^{-12}\ \mathrm{F/m}$

The change δL in span due to the arc length constraint for an axially unconstrained flexure of initial length L is given by:

$$\delta L \approx -\frac{3u^2}{5L}$$

where u is the lateral deflection.

The fractional resistance change of a <110>-aligned piezoresistor subject to longitudinal and transverse stresses σ_L and σ_T may be written as:

$$\frac{\Delta R}{R} = \sigma_L \pi_L + \sigma_T \pi_T$$

where $\pi_L = \frac{1}{2}\{\pi_{11} + \pi_{12} + \pi_{44}\}$ and $\pi_T = \frac{1}{2}\{\pi_{11} + \pi_{12} - \pi_{44}\}$, and π_{ij} are the standard piezoresistive coefficients.

The maximum stress σ_{max} in a square membrane of side L and thickness h subject to a uniformly distributed load p is given by:

$$\sigma_{max} \approx 0.3\left(\frac{L}{h}\right)^2 p$$

The critical buckling load for a beam that is built-in at both ends is given by:

$$P_c = \frac{4\pi^2 EI}{L^2}$$

The lowest order self-resonance frequency of a cantilever is given by:

$$\omega_0 = \frac{3.52}{L^2}\sqrt{\frac{EI}{\rho A}}$$

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This question is compulsory.

1. a) List the four main transductions mechanisms used in MEMS sensors and briefly describe their underlying principles. Which two mechanisms are most widely used, and why? [6]

b) Show from the geometry of the unit cell that the sidewalls of an anisotropically etched cavity in a (100)-oriented silicon wafer are expected to form an angle of 54.7° with the wafer surface. Also show that vertical sidewalls can be obtained by anisotropically etching a (110)-oriented silicon wafer. What is the limiting shape of the opening at the wafer surface for each of these wafer orientations? [6]

c) By considering the expression for the stiffness of either a cantilever or a flexure, derive a scaling law for the elastic force. Also give one example of a force that has a lower scaling index than the elastic force, and one example of a force with a higher scaling index. In both cases comment on the implications of this for MEMS devices. [8]

d) A series of straight beams, built-in at both ends, is to be used to monitor the residual stress of a polysilicon deposition process. The beam lengths are chosen to be 2460, 1740, 1420, 1230 and 1100 μm. If the polysilicon layer thickness is 3 μm, and the two longest beams are found to be buckled after release, what can you say about the residual stress level? [6]

e) A resonant chemical sensor is fabricated from a silicon cantilever with a selectively absorbing polymer layer of thickness 0.5 μm covering its upper surface. The polymer has a density of 1050 kg/m^3. If the cantilever is 15 μm wide, 300 μm long, and 5 μm deep, calculate the nominal frequency of the lowest order vibrational mode. Also, if the smallest frequency shift the readout electronics can detect is 0.5 Hz, what will be the minimum detectable mass of analyte? [6]

Assume values of 170 GPa and 2330 kg/m^3 for the Young's modulus and density of silicon. You may assume that the Young's modulus of the polymer is negligible.

Question 1 continues on the next page

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Question 1 continued

f) Explain why it can be advantageous to integrate a MEMS device and its associated electronics on the same silicon substrate? Figure 1.1 shows a cross-section of a MEMS device with integrated electronics, fabricated by MEMS-first processing. Identify the materials labelled A to E. What are the advantages of MEMS-first processing over other integration approaches? [8]

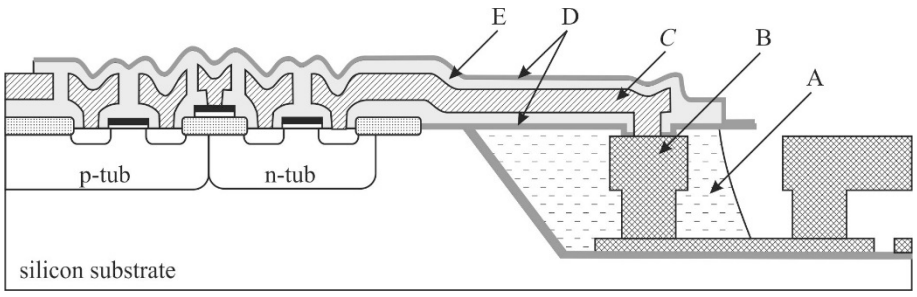


Figure 1.1

End of Question 1

Answer ANY TWO of the remaining questions (Questions 2, 3 and 4).

2. a) Sketch a hammock suspension and explain why it is non-linear. Also explain briefly why non-linearity is generally undesirable in a mass-spring-damper system. [6]

b) Figure 2.1 below shows a suspension design consisting of a portal frame with an additional flexure in between the portal frame flexures. Explain why, if the width of the third flexure is chosen correctly, this flexure can remain stress-free (apart from bending stresses) and therefore act as a linear spring for y-axis displacements of the moving element. Show that the required width is:

$$w_f = \sqrt[3]{2}\left(\frac{l_f}{l_p}\right)^{\frac{5}{6}} w_p$$

[10]

c) When a suspension of the type shown in Figure 2.1 is deflected in the y-direction, the portal frame flexures are subject to axial loads of:

$$\pm\frac{(l_f - l_p)}{2s}P$$

where P is the applied transverse load. By considering the equilibrium of the portal frame cross-bar, explain why these axial loads arise. [4]

Would you expect the axial loads in the flexures to affect the overall linearity of the suspension? Explain your reasoning. [4]

d) Four suspensions of the type in Figure 2.1 are used to support the proof mass of an in-plane silicon MEMS resonator as shown in Figure 2.2. The flexure lengths are l_p = 200 μm, l_f = 250 μm and the proof mass has an area of 0.72 mm^2. If the device is to have a resonance frequency of 10 kHz, and the suspensions are to be designed for linear operation, calculate the required flexure widths, w_p and w_f. [6]

Assume values of 170 GPa and 2330 kg/m^3 for the Young's modulus and density of silicon.

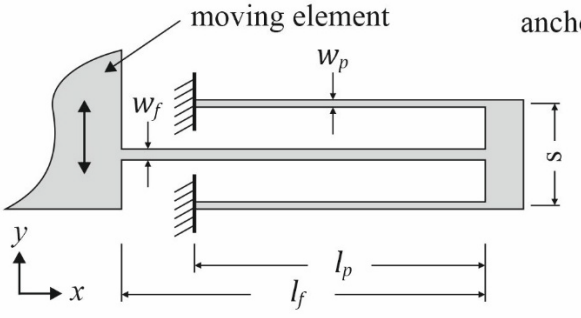


Figure 2.1

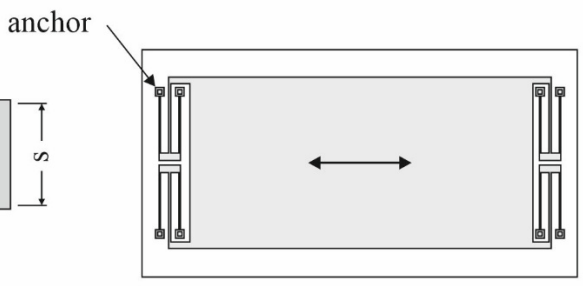


Figure 2.2

3. a) Starting from the generalised Hooke's law equations, show that the axial stress-strain relationships in a material where the z-component of axial stress is zero can be expressed in the form:

$$\sigma_x = \frac{E}{1-\upsilon^2}\varepsilon_x + \upsilon\frac{E}{1-\upsilon^2}\varepsilon_y$$

$$\sigma_y = \frac{E}{1-\upsilon^2}\varepsilon_y + \upsilon\frac{E}{1-\upsilon^2}\varepsilon_x$$

[6]

b) Using the above equations, show that the axial stresses at the edge of an xy-aligned rectangular membrane subject to a normal load will satisfy the relation:

$$\sigma_p = \nu\sigma_n$$

where σ_p and σ_n are the stresses parallel- and normal to the edge, respectively. [4]

c) Sketch the layout of a bulk micromachined, bridge-type piezoresistive pressure sensor. Why is the membrane in such a sensor aligned to the <110> directions? [6]

d) For the device you have sketched in your answer to part c), derive an expression for the bridge output voltage in terms of the maximum membrane stress. [6]

e) You are tasked with designing piezoresistive pressure sensor that has a <110>-aligned square membrane of side length 1 mm and a sensitivity of 200 μV/kPa at a bridge voltage of 5 V. What thickness of membrane should you choose? [6]

If maximum membrane stress in your sensor is to be limited to 200 MPa, what will be the full-scale measurement range? [2]

Assume the following values for the piezoresistive coefficients of the silicon (all in units of 10^{-11} Pa^{-1}): $\pi_{11} = 6.6$, $\pi_{12} = -1.1$, $\pi_{44} = 138.1$. Poisson's ratio for the relevant directions is $\nu = 0.064$.

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4. a) Figure 4.1 shows the electromechanical model for an out-of-plane electrostatic actuator, comprising a suspended moveable plate and a fixed plate covered by a dielectric layer. Derive an expression for the total force acting on the moveable plate and sketch the variation of this force with the gap g for several values of V in order to illustrate the origin of the *snap-down* instability. Also sketch the variation of g with applied voltage. [8]

b) State the conditions that apply at the point of snap-down, and hence show that the snap-down voltage is given by:

$$V_P = \sqrt{\frac{8k(g_0 + t_d/\varepsilon_r)^3}{27\varepsilon_0 A}}$$ [8]

c) Assuming the moveable plate has snapped down, to what level will the applied voltage need to be reduced before the moveable plate will spring back up again? [4]

d) Figure 4.2 shows an electrostatically actuated, capacitive shunt RF MEMS switch comprising a gold bridge suspended over a transmission line. The bridge is 500 μm long, 100 μm wide and 1 μm thick, and the signal line beneath is 100 μm wide. The dielectric spacer is a 100 nm-thick layer of silicon nitride. The initial gap between the bridge and the dielectric is 3 μm.

Calculate the stiffness of the gold bridge, and hence estimate the pull-down voltage, assuming that the electrostatic force is a point load at the centre of the bridge. [6]

e) What advantages do RF MEMS switches of the kind shown in Figure 4.2 offer over more traditional solid state devices? [4]

You should assume E = 80 GPa for gold, and ε_r = 7.6 for silicon nitride.

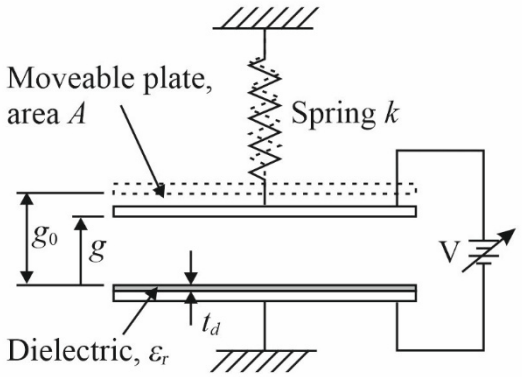


Figure 4.1

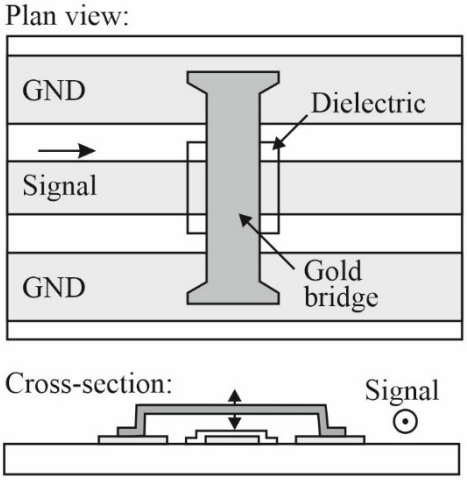


Figure 4.2